

5.1.4 Components of Neural Networks

نیورل نیٹ ورک کے اجزاء ہیں: ان پٹ تہ، نیورون، پوشیدہ تہ، وزن (Weights)، بائیس (Bias)، ایکٹیویشن فنکشن، آؤٹ پٹ تہ اور لاس فنکشن۔

CS XII

Chapter 5: Application and Impacts of Computing

Lecture 05 of 17

★ Golden

5.1.4 Components of Neural Networks ★ Golden

LEARN IT

The components of a neural network can be understood through its **layered architecture**. Data **enters**, gets **processed**, and **exits**, with mathematical adjustments happening inside.

#	Component	Description
1	Input layer	Receives the initial raw data (for example image pixels, sound waves, text).
2	Neurons / nodes	Small processing units that receive, calculate, and pass information forward.
3	Hidden layer(s)	The layers between input and output. They process the data and learn patterns from it.
4	Weights	Numbers that show the importance or strength of connections between neurons.
5	Bias	An extra constant value added to help the model adjust its result and improve accuracy.
6	Activation function	A mathematical gate that decides whether the information is important enough to move to the next layer.
7	Output layer	Gives the final computed result, decision or prediction (for example "This is a cat").
8	Loss function	Checks the error (how much the model's answer is wrong compared with reality) so the network can correct itself.

What happens inside one neuron

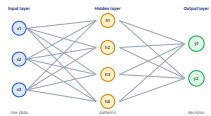
Each neuron does a short calculation:

- Multiply each input x_i by its weight w_i .
- Add those products together, then add the **bias** b . This total is the **weighted sum** s .
- Pass s through an **activation function**. A simple exam-friendly gate is a **threshold (step) function**: if s is at least the threshold, the neuron **fires** (output = 1); otherwise output = 0.

In symbols: $s = (x_1 \times w_1) + (x_2 \times w_2) + \dots + b$, then $y = 1$ if $s \geq \text{threshold}$, else $y = 0$.



One neuron: inputs $\times$ weights are summed, bias is added, then an activation function decides the output y .



A small neural network: 3 input nodes, 4 hidden nodes, 2 output nodes. Each line is a weighted connection.

WORKED EXAMPLE

Worked example: neuron calculation with numbers. A college in Larkana uses a tiny network to guess whether a student is ready for a test. One hidden neuron receives two inputs:

Symbol	Meaning	Value
x_1	Hours studied yesterday (scaled 0 to 1)	0.8
x_2	Attendance this month (scaled 0 to 1)	0.5
w_1	Weight of hours studied	0.4
w_2	Weight of attendance	0.6
b	Bias	0.1
threshold	Activation gate	0.5

Step 1 — weighted products.

$$x_1 \times w_1 = 0.8 \times 0.4 = \mathbf{0.32}$$

$$x_2 \times w_2 = 0.5 \times 0.6 = \mathbf{0.30}$$

Step 2 — add bias to get the weighted sum s .

$$s = 0.32 + 0.30 + 0.1 = \mathbf{0.72}$$

Step 3 — activation (threshold 0.5). Is $0.72 \geq 0.5$? **Yes**, so the neuron fires: $y = 1$ (ready for the test).

Step 4 — a second student with $x_1 = 0.2$ (almost no study) and $x_2 = 0.3$:

$$s = (0.2 \times 0.4) + (0.3 \times 0.6) + 0.1 = 0.08 + 0.18 + 0.1 = \mathbf{0.36}$$

0.36 is **less than** 0.5, so $y = 0$ (not ready). The same neuron, different inputs, different decision.

You can check the first student in Python 3:

```
x1, x2 = 0.8, 0.5
w1, w2 = 0.4, 0.6
b = 0.1
s = x1 * w1 + x2 * w2 + b
y = 1 if s >= 0.5 else 0
print(round(s, 2))
print(y)
```

```
OUTPUT
0.72
1
```

Loss function, briefly. If the true answer for the first student was actually "not ready" (true $y = 0$) but the model said 1, the loss is high. Training then slightly changes weights and bias so the next guess is closer to reality.

REMEMBER IT

Weights = importance. **Bias = extra nudge.** **Activation = yes/no gate.** **Loss = how wrong.** A large weight means that input matters more. Bias shifts the sum up or down so the gate can open at the right place.

COMMON MISTAKE

Mistake: Forgetting to add the bias, or multiplying instead of adding after the products.

Fix: Always **multiply first** (each x by its w), **then add** the products, **then add b** , **then** apply the activation. Write the four lines in the exam as we did above.

آرڈو میں سمجھیں

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EXAM TIP

Golden questions: "List the components of a neural network" (often 8 names + one line each), "What are weights and bias?", and a **numerical neuron**: given x , w , b and a threshold, find s and y .

Show every arithmetic step. Underline the final y . If asked for a diagram, draw Input $\rightarrow$ Hidden $\rightarrow$ Output with circles and connecting lines, and label weights on the lines. Mention the loss function as the component that measures error so the network can correct itself.

CHECK YOURSELF

- What do weights show?
- Using $x_1 = 1$, $x_2 = 0$, $w_1 = 0.5$, $w_2 = 0.5$, $b = 0$, threshold = 0.4, what is y ?
- What does the loss function check?

Answers: 1. The importance or strength of connections between neurons. 2. $s = 1 \times 0.5 + 0 \times 0.5 + 0 = 0.5$; $0.5 \geq 0.4$ so $y = 1$. 3. The error — how much the model's answer is wrong compared with reality.

Next lecture: 5.1.5 Components of Deep Learning